

Performance Testing Report

Date	15 July 2026
Team ID	XXXXXX
Project Name	StudyAI – Smart AI educate Companion & Quiz Generator
Maximum Marks	5 Marks

Performance Testing Report:

Step 1: Test Scenarios & Tooling

S.No	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	<i>Simulating 100 concurrent users logging in and generating summaries simultaneously</i>	<i>Apache JMeter, Postman</i>
2	Stress Testing	<i>Uploading large PDF files and generating quizzes continuously until system slowdown</i>	<i>Apache JMeter</i>
3	Spike Testing	<i>Sudden burst of 200 users generating flashcards at the same time</i>	<i>JMeter</i>
4	Endurance Testing	<i>Running document upload and AI generation requests continuously for 6 hours to check memory leaks and stability</i>	<i>JMeter, Browser Developer Tools</i>

Step 2: Key Performance Indicators (KPIs) & Results

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
1	Average Response Time	<i>< 2 seconds</i>	<i>1.4 seconds</i>	Pass
2	Peak Response Time	<i>< 5 seconds</i>	<i>3.8 seconds</i>	Pass
3	Throughput (Requests/sec)	<i>> 50 RPS</i>	<i>65 RPS</i>	Pass
4	Error Rate	<i>< 2%</i>	<i>1.1%</i>	Pass
5	CPU/Memory Utilization	<i>< 80% under load</i>	<i>CPU: 65%, Memory: 70%</i>	Pass